



CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

3443 Routier Road, Suite A
Sacramento, CA 95827-3003
Phone (916) 255-3000
FAX (916) 255-3015

Cal/EPA



Pete Wilson, Governor

11 December 1997

CERTIFIED MAIL
Z 684 995 564

Steve Brodnansky
Lake County Sanitation District - Special Districts
Courthouse
255 N. Forbes Street
Lakeport, CA 95453

NOTICE OF ADOPTION
OF
REVISION WASTE DISCHARGE REQUIREMENTS FOR
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

Waste Discharge Requirements Order No. 97-249 for the above named discharger was adopted by the California Regional Water Quality Control Board, Central Valley Region, on 5 December 1997.

JACK E. DEL CONTE
Supervising Engineer

GAI:dlk

Enclosures (Adopted Order and Standard Provisions)

cc+w/o encl: Office of Drinking Water, Department of Health Services, Sacramento
Environmental Management Branch, Department of Health Services, Sacramento
Department of Fish and Game, Region III, Yountville
Ms. Frances McChesney, Office of Chief Counsel, State Water Resources Control
Board, Sacramento
Division of Water Quality, State Water Resources Control Board, Sacramento
Mr. John Herren, Division of Clean Water Programs, State Water Resources
Control Board, Sacramento
Mr. Martin Winston, Lake County, Health Department, Lakeport
Lake County Planning Department, Lakeport

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGIONAL

ORDER NO. 97-249

WASTE DISCHARGE REQUIREMENTS
FOR
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The Lake County Sanitation District, Assessment District No. 2-2 (hereafter Discharger) submitted a revised Report of Waste Discharge, dated 26 September 1997, and a Corrective Action Plan, dated 22 September 1997, for its Middletown Wastewater Treatment Facility in Lake County (Assessor's Parcel No. 13-011-54).
2. Waste Discharge Requirements Order No. 88-175, adopted by the Board on 23 September 1988, prescribes requirements for a discharge from the community of Middletown to stabilization ponds and irrigation of fodder crops.
3. Order No. 88-175 is neither adequate nor consistent with current plans and policies of the Board.
4. The Discharger proposes to treat approximately 0.15 mgd average daily dry weather flow and a peak wet weather flow of approximately 0.5 mgd of domestic sewage from the community of Middletown with treatment ponds. Treated effluent will be disposed of primarily through the Geysers effluent pipeline by injection into the Southeast Geysers steam field. The existing spray irrigation system will be used for effluent disposal by irrigating fodder crops when required.
5. The Lake County Sanitation District, Assessment District 2-2, is in the Northwest quadrant of Collayomi Valley in the NE 1/4 of Section 32, T11N, R7, MDB&M, with surface water drainage to Putah Creek, as shown in Attachment A, which is attached hereto and part of the Order by reference.
6. The beneficial uses of Putah Creek are municipal, industrial, and agricultural supply; recreation; esthetic enjoyment; ground water recharge; hydroelectric power generation; and preservation and enhancement of fish, wildlife and other aquatic resources.
7. The beneficial uses of the ground water are municipal, industrial, and agricultural supply.
8. The Board adopted a Water Quality Control Plan, Third Edition, for the Sacramento River Basin and San Joaquin River Basin (hereafter Basin Plan), which contains water quality objectives. These requirements are consistent with that Plan.

WASTE DISCHARGE REQUIREMENTS
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

-2-

9. The County of Lake has adopted a final Environmental Impact Report (EIR), in accordance with the California Environmental Quality Act (Public Resources Code Section 21000, et seq.), and the State Guidelines. The project may have the following significant impacts on water quality:
 - a. Potential for obnoxious odor;
 - b. Potential problems of flooding of the treatment and disposal facilities, raw sewage spills, and ground water pollution;
 - c. Potential for damage to water quality and aquatic life from sewer force main rupture; and
 - d. Health hazard from spray irrigation.
10. The Board has reviewed the EIR, and compliance with the waste discharge requirements will avoid significant impacts on water quality listed in finding No. 9 or there are overriding considerations as follows:
 - a. Odors will be mitigated by relative isolation (1,000 feet from the nearest residence), and by proper design, maintenance, and operation of the treatment units and sludge handling facilities, as required by Discharge Specifications B.1 and B.9. These measures include design review by the State Water Resources Control Board, Loans and Grants Branch, artificial aeration and addition of chemicals, such as lime as necessary. Proper operation and maintenance will be checked by periodic inspection by Board staff.
 - b. Flooding, raw sewage spills, and ground water pollution will be mitigated by requiring construction of the treatment and disposal facilities above the floodplain caused by a 100-year storm event, installation of a standby pump and electric generator, and lining of the treatment, holding, and storage ponds, as required by Discharge Prohibitions A.1, 2, 3, and 4, and Discharge Specifications B.1 through 12, inclusive.
 - c. Damage to water quality and aquatic life from sewer main rupture will be mitigated by proper design (reviewed by the State Water Resources Control Board, Loans and Grants Branch), installation and maintenance, as required by Discharge Prohibitions A.1 and 2, and Discharge Specifications B.1, 2, and 6. The potential for force main rupture can be minimized, but may not be substantially mitigated. However, benefits of installing a sewer system outweigh the risk of a force main rupture. Absent a sewer system, the community of Middletown will continue to use on-site sewage systems. These systems may be currently degrading ground water and threatening the public health. Therefore, the risk of a force main rupture is acceptable.

WASTE DISCHARGE REQUIREMENTS
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

-3-

- d. Health hazards from spray irrigation will be mitigated by disinfection, as required by Discharge Specification B.3, 4, and 8.
11. The installation of a sewerage system will be a positive impact in that it will mitigate the existing pollution of ground water from individual on-site septic tank/leachfield systems.
12. The action to adopt waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality Act in accordance with Section 15301, Title 14, California Code of Regulations (CCR).
13. The discharge is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq., (hereafter Title 27). The exemption, pursuant to Section 20090(b), is based on the following:
 - a. The Board is issuing waste discharge requirements
 - b. The discharge complies with the Basin Plan, and,
 - c. The wastewater does not need to be managed according to 22 CCR, Division 4.5, Chapter 11, as a hazardous waste.
14. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge.
15. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 88-175 is rescinded and Lake County Sanitation District, Assessment District No. 2-2, its agents, successors and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. The direct discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. The by-pass or overflow of untreated or partially treated waste is prohibited.
3. The application of effluent on areas which do not have vegetative cover or other erosion control measures in effect is prohibited.

WASTE DISCHARGE REQUIREMENTS
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

-4-

4. The discharge of wastewater to spray fields within 24 hours of or during rainfall is prohibited.

B. Discharge Specifications:

1. Neither the treatment nor the discharge shall cause a pollution or nuisance as defined by the California Water Code, Section 13050.
2. The discharge shall not cause degradation of any water supply.
3. When irrigating, the discharge shall remain within the designated disposal area at all times.
4. The discharge of an effluent to the storage reservoir in excess of the following limits is prohibited:

<u>Constituents</u>	<u>Units</u>	<u>30-Day Average</u>	<u>Monthly Median</u>	<u>Daily Maximum</u>
BOD ¹	mg/l	40	---	80
Total Coliform	MPN/100 ml	---	23	230

¹ 5-day, 20°C biochemical oxygen demand

5. During irrigation, any runoff from irrigated areas shall be returned to the storage reservoir or appropriate irrigation block.
6. The discharge shall not have a pH less than 6.5 nor greater than 8.5.
7. The 30-day average daily dry weather discharge flow shall not exceed 0.15 million gallons.
8. Reclaimed wastewater shall meet the criteria contained in Title 22, Division 4, California Code of Regulations (Section 60301, et seq.).
9. The dissolved oxygen content of treatment ponds shall not be less than 1.0 mg/l for 16 hours in any 24-hour period.
10. The invert of all treatment ponds shall have liner material such that the permeability does not exceed 1×10^{-6} cm/sec.
11. The ground surface of the irrigation area shall be a minimum of five feet above the high ground water elevation during the periods effluent is being applied.

12. All stabilization, holding, or storage ponds, and disposal areas shall be protected from wash-out by a 100-year storm event.

C. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Title 27 CCR, Division 2, Subdivision 1 and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice from a previously approved practice shall be reported to the Executive Officer and U.S. Environmental Protection Agency (EPA) Regional Administrator at least **90 days** in advance of the change.
3. Use and disposal of sewage sludge shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.

If the State Water Resources Control Board and the Regional Water Quality Control Boards are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

4. The Discharger is encouraged to comply with the State Guidance Manual issued by the Department of Health Services, titled "*Manual of Good Practice for Landscaping of Sewage Sludge*".
5. The Discharger shall submit a sludge disposal plan describing the annual volume of sludge generated by the plant and specifying the disposal practices.

D. Ground Water Limitations:

The discharge shall not cause underlying ground water to be degraded.

E. Provisions:

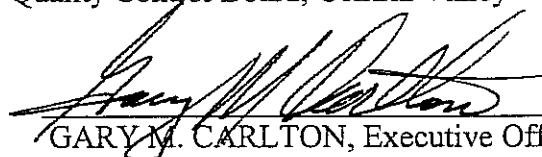
1. The Discharger shall comply with Monitoring and Reporting Program No. 97-249, which is a part of this Order, and any revisions thereto as ordered by the Executive Officer.

WASTE DISCHARGE REQUIREMENTS
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

-6-

2. The Discharger shall comply with all the items of the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are referred to as "Standard Provision(s)".
3. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
4. At least **90 days** prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
5. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
6. The Discharger shall provide certified wastewater treatment plant operators in accordance with regulations adopted by the State Water Resources Control Board.
7. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
8. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with Title 22 CCR, Division 4.
9. The Board will review this Order periodically and may revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 5 December 1997.


GARY M. CARLTON, Executive Officer

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

REVISED MONITORING AND REPORTING PROGRAM NO. 97-249

FOR
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

This monitoring and reporting program (MRP) incorporates requirements for monitoring of the wastewater treatment plant. This MRP is issued pursuant to Water Code Section 13267. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Executive Officer.

All wastewater samples should be representative of the volume and nature of the discharge. The time, date, and location of each grab sample shall be recorded on the chain of custody form. Process wastewater flow monitoring shall be conducted continuously using a flow meter and shall be reported in cumulative gallons per day.

Field test instruments (such as pH and dissolved oxygen) may be used provided that:

1. The operator is trained in the proper use of the instrument;
2. The instruments are field calibrated at the frequency recommended by the instrument manufacturer or at the frequency recommended by an industry standard;
3. Instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
4. Field calibration reports are submitted as described in the "Reporting" section of this MRP.

INFLUENT MONITORING

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	gpd	Continuous	Daily	Monthly
BOD ₅ ¹	mg/L	Grab	Monthly	Monthly

¹ 5-day, 20°C Biochemical Oxygen Demand.

EFFLUENT MONITORING

Effluent samples shall be collected downstream from the last treatment pond, and prior to discharge to the effluent storage pond. Effluent monitoring shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Total Coliform Organisms ¹	MPN/100 mL ²	Grab	Weekly ³	Monthly
Chlorine Residual	mg/L	Grab	Weekly ³	Monthly

REVISED MONITORING AND REPORTING PROGRAM NO. 97-249
 LAKE COUNTY SANITATION DISTRICT
 ASSESSMENT DISTRICT NO. 2-2
 MIDDLETOWN WASTEWATER TREATMENT FACILITY
 LAKE COUNTY

-2-

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
BOD ₅ ⁴	mg/L	Grab	Twice Monthly	Monthly
Settleable Solids	mg/L	Grab	Monthly	Monthly
Total Dissolved Solids	mg/L	Grab	Monthly	Monthly
Nitrate as Nitrogen	mg/L	Grab	Monthly	Monthly
Total Kjeldahl Nitrogen	mg/L	Grab	Monthly	Monthly
Specific Conductivity@ 25°C	µmhos/cm	Grab	Monthly	Monthly
Standard Minerals ⁵	mg/L	Grab	Annually	Annually

¹ Using a minimum of 10 tubes or two dilutions.

² Most probable number per 100 ml.

³ During irrigating of the land application area.

⁴ 5-day, 20°C Biochemical Oxygen Demand.

⁵ Standard Minerals shall include the following: boron, calcium, iron, manganese, magnesium, potassium, sodium, chloride, total alkalinity (including alkalinity series), and hardness.

POND MONITORING

Each of the wastewater treatment ponds and the effluent storage pond shall be monitored for the parameters specified below.

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Dissolved Oxygen ^{1,4}	mg/L	Grab	Weekly	Monthly
pH ⁴	pH units	Grab	Weekly	Monthly
Freeboard	0.25 feet	Observation	Weekly	Monthly
Berm Seepage ²	NA	Observation	Weekly	Monthly
Odors ³	--	Observation	Weekly	Monthly

¹ Samples shall be collected at a depth of one foot from each pond in use, opposite the inlet. Samples shall be collected between 0700 and 0900 hours.

² Reservoir containment levees shall be observed for signs of seepage or surfacing water along the exterior toe of the levees and dam. If surfacing water is found, then a sample shall be collected and tested for total coliform organisms and total dissolved solids.

³ The presence of strong or unusual odors shall be reported.

⁴ Hand held meter may be used.

LAND APPLICATION AREA MONITORING

Monitoring of the land application area shall be conducted **daily** during irrigating and the results shall be included in the monthly monitoring report. Evidence of erosion, saturation, irrigation runoff, or the presence of nuisance conditions shall be noted in the report. Effluent monitoring results shall be used in calculations to ascertain loading rates at the application area. Monitoring of the land application area shall include the following:

REVISED MONITORING AND REPORTING PROGRAM NO. 97-249
 LAKE COUNTY SANITATION DISTRICT
 ASSESSMENT DISTRICT NO. 2-2
 MIDDLETOWN WASTEWATER TREATMENT FACILITY
 LAKE COUNTY

-3-

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	Gallons	Continuous	Daily	Monthly
Local Rainfall	Inches	Observation	Daily	Monthly
Acreage Applied ¹	Acres	Calculated	Daily	Monthly
Application Rate ²	gal/acre/day	Calculated	Daily	Monthly
BOD ₅ Loading Rate ²	lbs/acre/day ³	Calculated	Monthly	Monthly
Total Nitrogen Loading Rate ²	lbs/acre/month ⁴	Calculated	Monthly	Monthly
Total Dissolved Solids Loading Rate ²	lbs/acre/month ⁴	Calculated	Monthly	Monthly

¹ Land application areas shall be identified.

² For each land application area.

³ Report 7-day average and maximum daily loading.

⁴ Report monthly total and cumulative annual to date.

The entire irrigated area shall be periodically inspected during or immediately following an irrigation event to identify any equipment malfunction or other circumstances that might allow irrigation runoff to leave the irrigation area and/or create ponding conditions that violate the Waste Discharge Requirements. A daily log of these inspections shall be kept at the facility and made available for review upon request.

SLUDGE MONITORING

In accordance with EPA's POTW Sludge Sampling and Analysis Guidance Document, August 1989, a composite sample of sludge shall be collected when removed from the treatment and effluent storage ponds and tested for the following metals:

Cadmium	Copper	Nickel
Chromium	Lead	Zinc

Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis for part of the annual report.

GROUNDWATER MONITORING

The groundwater monitoring program shall begin in the fourth quarter of 2004. Prior to construction and/or sampling of any additional groundwater monitoring wells, the Discharger shall submit plans and specifications to the Regional Board for review and approval. Once installed, all new wells shall be

REVISED MONITORING AND REPORTING PROGRAM NO. 97-249
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

-4-

added to the MRP and shall be sampled and analyzed according to the schedule below. Existing wells to be monitored are MW-1, MW-2, MW-3, and MW-4.

Prior to sampling, the groundwater elevations shall be measured and each well shall be purged at least three well volumes until temperature, pH and electrical conductivity have stabilized. Depth to groundwater shall be measured to the nearest 0.01 feet. Samples shall be collected using standard EPA methods. Groundwater monitoring shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency⁴</u>	<u>Reporting Frequency⁴</u>
Depth to Groundwater	0.01 feet	Measurement	Quarterly	Quarterly
Groundwater Elevation ¹	0.01 feet	Calculated	Quarterly	Quarterly
Gradient	feet/feet	Calculated	Quarterly	Quarterly
Gradient Direction	Degrees	Calculated	Quarterly	Quarterly
pH ²	pH units	Grab	Quarterly	Quarterly
Specific Conductivity@ 25°C	µmhos/cm	Grab	Quarterly	Quarterly
Total Coliform Organisms	MPN/100 mL	Grab	Quarterly	Quarterly
Total Dissolved Solids	mg/L	Grab	Quarterly	Quarterly
Nitrate as Nitrogen	mg/L	Grab	Quarterly	Quarterly
Total Kjeldahl Nitrogen	mg/L	Grab	Quarterly	Quarterly
Standard Minerals ³	mg/L	Grab	Annually	Annually

¹ Groundwater elevation shall be determined based on depth-to-water measurements from a surveyed measuring point elevation on the well.

² Hand held field meter may be used.

³ Standard Minerals shall include the following: boron, calcium, iron, manganese, magnesium, potassium, sodium, chloride, total alkalinity (including alkalinity series), and hardness.

⁴ Beginning with the fourth quarter 2004.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., process wastewater effluent, groundwater well, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported in the next scheduled monitoring report.

As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all Quarterly Groundwater Monitoring Reports and the annual groundwater evaluation shall be prepared under the direct supervision of a Registered Engineer or Geologist and signed by the registered professional.

A. Monthly Monitoring Reports

Monthly reports shall be submitted to the Regional Board by the **1st day of the second month** following the end of the reporting period (i.e. the August monthly report is due by 1 October). Monthly reports for the months of March, June, September, and December may be submitted as part of the Quarterly Monitoring Report, if desired. The monthly reports shall include the following:

1. Results of influent, effluent, pond, and land application monitoring;
2. A comparison of monitoring data to the discharge specifications and an explanation of any violation of those requirements. Data shall be presented in tabular format;
3. If requested by staff, copies of laboratory analytical report(s); and
4. A calibration log verifying calibration of all hand held monitoring instruments and devices used to comply with the prescribed monitoring program;

B. Quarterly Monitoring Reports

Beginning with the fourth quarter of 2004, the Discharger shall establish a quarterly sampling schedule for groundwater monitoring such that samples are obtained approximately every three months. Quarterly monitoring reports shall be submitted to the Regional Board by the **1st day of the second month after the quarter** (i.e. the January-March quarterly report is due by May 1st) and may be combined with the monthly report. The Quarterly Report shall include the following:

1. Results of groundwater monitoring;
2. A narrative description of all preparatory, monitoring, sampling, and analytical testing activities. The narrative shall be sufficiently detailed to verify compliance with the WDRs, this MRP, and the Standard Provisions and Reporting Requirements. The narrative shall be supported by field logs for each well documenting depth to groundwater; parameters measured before, during, and after purging; method of purging; calculation of the casing volume; and total volume of water purged;
3. Calculation of groundwater elevations, an assessment of the groundwater flow direction and gradient on the date of measurement, comparison to previous flow direction and gradient data, and discussion of seasonal trends, if any;
4. A narrative discussion of the analytical results for all media and locations monitored, including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable);

REVISED MONITORING AND REPORTING PROGRAM NO. 97-249
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

-6-

5. A comparison of monitoring data to the discharge specifications, groundwater limitations, and surface water limitations, and explanation of any violation of those requirements;
6. Summary data tables of historical and current water table elevations and analytical results;
7. A scaled map showing relevant structures and features of the facility, the locations of monitoring wells and other sampling stations, and groundwater elevation contours referenced to mean sea level datum; and
8. Copies of laboratory analytical report(s) for groundwater monitoring.

C. Annual Monitoring Reports

An Annual Report shall be prepared as the fourth quarter monitoring report and shall include all monitoring data required in the monthly/quarterly schedule. The Annual Report shall be submitted to the Regional Board by **1 February of each year** and shall include the following:

1. The contents of the regular groundwater monitoring report for the last sampling event of the year;
2. If requested by staff, tabular and graphical summaries of all data collected during the year;
3. Data for monitoring the effluent, groundwater, and supply water performed on an annual basis;
4. An evaluation of the groundwater quality beneath the facility;
5. An evaluation of the performance of the wastewater treatment system, as well as a forecast of the flows anticipated in the next year;
6. Verification of appropriate employee training for all personnel involved in operation and maintenance of wastewater treatment system;
7. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements; and
8. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.
9. Results of a thorough pond liner inspection performed once yearly when the water level is the lowest. Describe all problems found, repairs made, repairs needed, and scheduled repair/replacement dates.

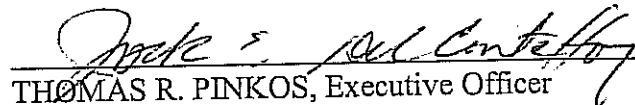
REVISED MONITORING AND REPORTING PROGRAM NO. 97-249
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

-7-

A letter transmitting the self-monitoring reports shall accompany each report. Such a letter shall include a discussion of requirement violations found during the reporting period, and actions taken or planned for correcting noted violations, such as operation or facility modifications. If the discharger has previously submitted a report describing corrective actions and/or a time schedule for implementing the corrective actions, reference to the previous correspondence will be satisfactory. The transmittal letter shall contain a statement by the discharger, or the discharger's authorized agent, under penalty of perjury, that to the best of the signer's knowledge the report is true, accurate and complete.

The Discharger shall implement the above monitoring program as of the date of this Order.

Ordered by:


THOMAS R. PINKOS, Executive Officer

9/21/04
(Date)

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY

MONITORING AND REPORTING PROGRAM NO. 97-249
FOR
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

Sample stations shall be established such that the samples are representative of the volume and nature of the discharge. Specific sample station locations shall be established under direction of the Board's staff, and a description of the stations shall be attached to this Order.

INFLUENT MONITORING

Samples shall be collected at approximately the same time as effluent samples and should be representative of the influent for the period sampled. The following shall constitute the influent monitoring program:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Flow	mgd	24 or 72-hour Composite	Flow estimated weekly

EFFLUENT DISCHARGE MONITORING

Effluent samples shall be collected downstream from the last treatment pond, and prior to discharge to the effluent storage pond. Effluent samples should be representative of the volume and nature of the discharge. Samples collected from the outlet structure of the treatment ponds will be considered adequately composited. The following shall constitute the effluent monitoring program:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sampling</u>	<u>Sampling Frequency</u>
20°C BOD ₅	mg/l	24 hr. Composite	Weekly
Settleable Matter	ml/l	Grab	Weekly
Specific Conductivity	µmhos/cm	Grab	Weekly
pH	pH Units	Grab	Weekly
Total Coliform	MPN/100 ml	Grab	Weekly when irrigating
Organisms			
Chlorine Residual	mg/l	Grab	Daily when irrigating

MONITORING AND REPORTING PROGRAM
LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

2-

If the discharge is intermittent rather than continuous, then on the first day of each such intermittent discharge, the Discharger shall monitor and record data for all of the constituents listed above, after which the frequencies of analysis given in the schedule shall apply for the duration of each such intermittent discharge. In no event shall the Discharger be required to monitor and record data more often than twice the frequencies listed in the schedule.

POND MONITORING

The wastewater treatment ponds' and storage reservoir's freeboard shall be recorded weekly and submitted to the Board. Recorded data shall be to the nearest half-foot.

These ponds shall also be inspected weekly for rodent damage. Observations shall be reported in each monthly monitoring report.

Treatment pond numbers one through four shall be tested every ten years or greater than fifty-year storm event, whichever occurs first, for compliance with the liner permeability requirement of 1×10^{-6} cm/sec. A proposed testing methodology shall be submitted to the Board for approval at least 90 days prior to the scheduled test date. The first test was conducted during 1993. Test results shall be submitted to the Board within one month after testing is complete.

GROUND WATER MONITORING

The monitoring wells Nos. MW-1, MW-2, MW-3, and MW-4 shall be monitored for the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Ground Water Elevation	Feet, MSL measured to nearest 0.01 foot	Grab	Monthly
Specific Conductivity	μ mhos/cm	Grab	Quarterly
Nitrate nitrogen (NO ₃) as NO ₃	mg/l	Grab	Quarterly

SOIL MONITORING

Annually, unless the disposal area has not been used for effluent disposal during the previous 12 months, a random soil sample shall be collected from the spray disposal area at a depth of one foot and analyzed for:

1. Specific Conductance @ 25°C
2. Nitrate nitrogen (NO₃) as NO₃

The sample location shall be referenced with 10 feet to the nearest Section Corner or Quarter Corner.

REPORTING

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly the compliance with waste discharge requirements.

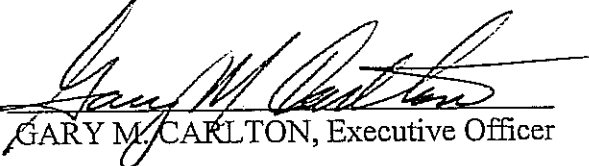
Monthly monitoring reports shall be submitted to the Regional Board by the 15th day of the following month.

The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Board.

Upon written request of the Board, the Discharger shall submit a report to the Board by 30 January of each year. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year. In addition, the Discharger shall discuss the compliance record and the corrective actions taken or planned which may be needed to bring the discharge into full compliance with the waste discharge requirements.

The Discharger shall implement the above monitoring program on the effective date of this Order.

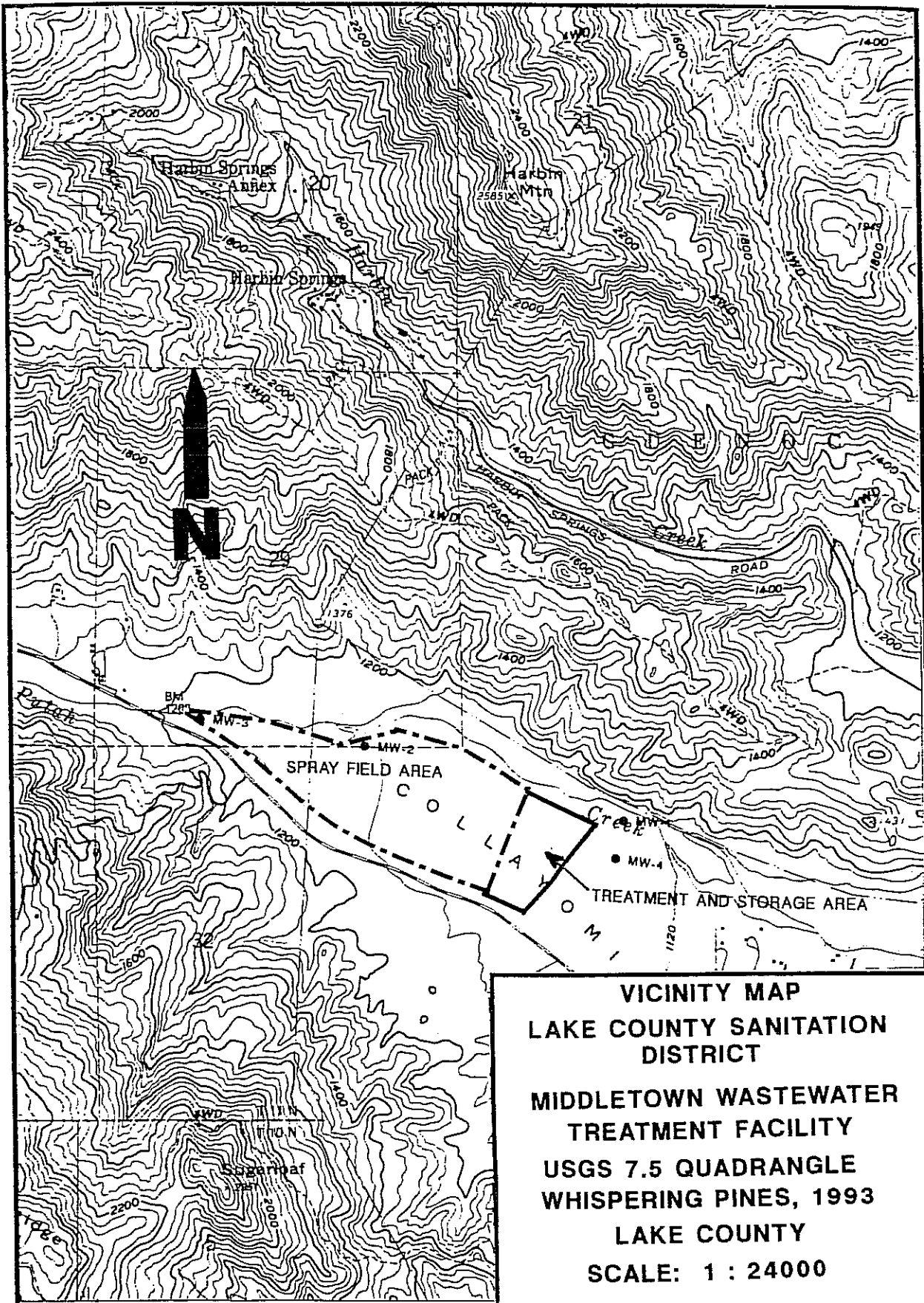
Ordered by:


GARY M. CARLTON, Executive Officer

5 December 1997

(Date)

ATTACHMENT A



INFORMATION SHEET

LAKE COUNTY SANITATION DISTRICT
ASSESSMENT DISTRICT NO. 2-2
MIDDLETOWN WASTEWATER TREATMENT FACILITY
LAKE COUNTY

This is an existing community of approximately 1,400 people. Soil in the area is very gravelly, sandy loam (Talmage). The Middletown Wastewater Treatment Facility has been subject to a number of pollution studies. The Lake County Sanitation District (LACOSAN) proposes to make the following changes to the facility:

- 1. Collection System:** Over the next three years, identify and repair collection lines and manholes to reduce infiltration/inflow into the collection system. This involves the sealing of approximately 25 manholes and 10 collection line spot repairs.
- 2. Installation of Liners in Treatment Ponds 3 & 4:** This work will include raising the hydraulic gradient by one foot in all four treatment ponds and is intended to isolate the ponds from the influence of groundwater. A synthetic liner will be placed in Ponds 3 and 4. This will consist of a subdrain system and one foot of dirt cover over the bottom of the synthetic liner being placed in each of two ponds.
- 3. Limited Use of the Effluent Storage Pond (ESP):** A divider dike was placed in the ESP in 1996. This has resulted in the ESP meeting the plans and specifications. The long term use of this reservoir will be limited to emergency storage of effluent during periods of Geysers Pipeline dewatering for maintenance and repair. In addition, the ESP will be available of wet weather events, if needed.
- 4. Geysers Pipeline Disposal:** Effluent (from the Southeast Regional Wastewater Treatment Plant and the Middletown Wastewater Treatment Plant) mixed with make-up water from Clear Lake will be piped to the Southeast Geysers area from injection and geothermal reservoir enhancement. This is considered a long term disposal solution as well as an example of long term sustainable energy development. The pipeline also has significant designed capacity to accommodate future connections. The existing sprayfield would also be retained to provide addition operational flexibility.

The implementation of these changes and the addition of the Geysers Pipeline for storage and disposal will result in positive impacts to the operations of the treatment facility. These waste discharge requirements have been revised to reflect the aforementioned changes. Surface water drainage is to Putah Creek and Lake Berryessa.

